

ESAT CAMP MOCK TEST A

Physics

EC-PHY-A

PHYSICS

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.

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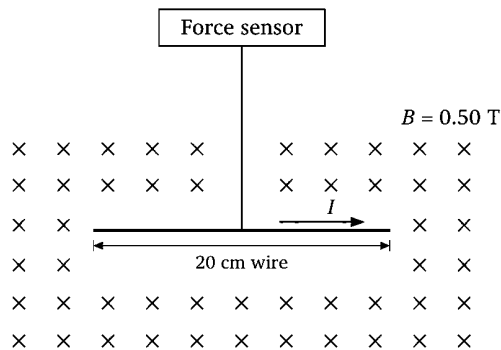
Physics

- 1 A point on a guitar string oscillates in simple harmonic motion with a frequency of 50 Hz and an amplitude of 4.0 mm.

What is the total distance travelled by this point during a time interval of 3.0 s?

- A 0.0 m
- B 0.6 m
- C 1.2 m
- D 2.4 m
- E 3.8 m

- 2 A straight, horizontal wire of length 20 cm is suspended from a sensitive force sensor. The wire is in a uniform magnetic field of strength 0.50 T, directed perpendicular to the wire. A total charge of 12 C passes through the wire over a time of 2.0 s. What is the magnitude of the change in the reading on the force sensor?



- A 0.20 N
- B 0.60 N
- C 1.2 N
- D 2.4 N
- E 3.0 N
- F 60 N

- 3 Two objects, X and Y, are in motion. Object X has 3 times the momentum of object Y and 4 times the kinetic energy of object Y.

What is the ratio of their masses, $\frac{m_X}{m_Y}$?

- A $\frac{3}{16}$
- B $\frac{3}{4}$
- C $\frac{4}{9}$
- D $\frac{9}{4}$
- E 12
- F 36

- 4 The intensity I of a physical field varies with distance r from a point source according to a power law relationship $I = kr^n$, where k and n are constants. Experimental measurements yield the following data:

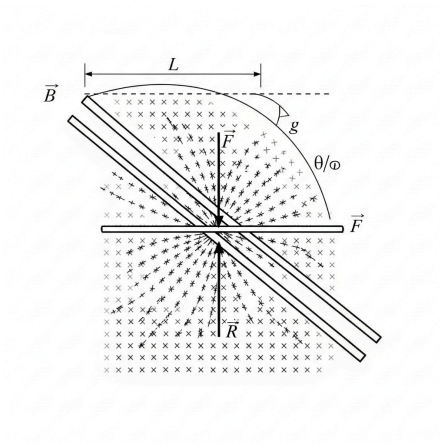
$$\text{At } r = 2.0 \text{ m, } I = 72 \text{ W m}^{-2}$$

$$\text{At } r = 4.0 \text{ m, } I = 18 \text{ W m}^{-2}$$

What is the intensity I at a distance of $r = 3.0 \text{ m}$?

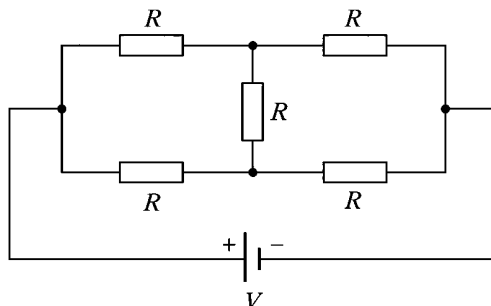
- A 24 W m^{-2}
- B 32 W m^{-2}
- C 36 W m^{-2}
- D 45 W m^{-2}
- E 48 W m^{-2}

- 5 A conducting rod slides without friction down two parallel conducting rails, which are inclined at a fixed angle to the horizontal. The entire setup is in a uniform magnetic field, perpendicular to the plane of the rails. The rod falls and reaches a terminal velocity, at which point the power dissipated as heat in the rod is P . The experiment is repeated, but with the magnetic field strength doubled. What is the new power dissipated in the rod when it reaches its new terminal velocity?

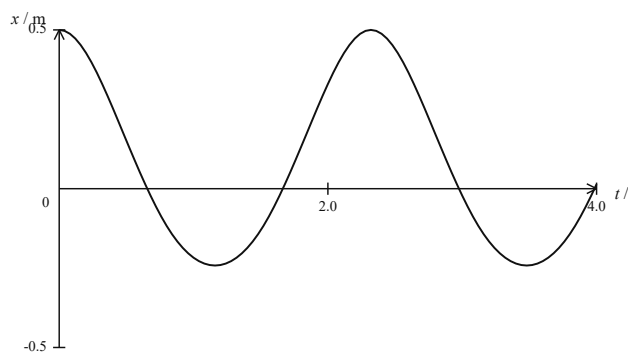


- A $P/16$
 B $P/4$
 C $P/2$
 D P
 E $2P$
 F $4P$
- 6 Two identical objects are heated from the same initial temperature to the same final temperature. One object is heated by a 20 W heater. The other is heated by a 30 W heater. The process takes 50 s longer with the 20 W heater than with the 30 W heater. What is the total thermal energy absorbed by each object?
- A 0.25 kJ
 B 0.50 kJ
 C 0.60 kJ
 D 1.5 kJ
 E 3.0 kJ
 F 6.0 kJ

- 7 A circuit is constructed with five identical resistors, each of resistance R , connected to a DC supply of voltage V . The circuit consists of two parallel branches, each containing two resistors in series. A fifth identical resistor connects the midpoint of the first branch to the midpoint of the second branch. The total power dissipated by the circuit is P . If the connecting resistor between the two branches is removed, which expression gives the new total power dissipated?



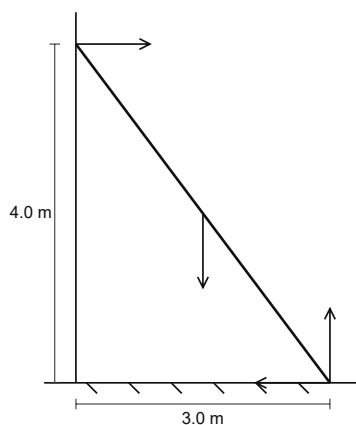
- A $\frac{1}{2}P$
 B $\frac{4}{5}P$
 C P
 D $\frac{5}{4}P$
 E $2P$
- 8 The graph shows the displacement of a particle undergoing simple harmonic motion as a function of time.



What is the average speed of the particle during the first 4.0 s of its motion?

- A 0 ms^{-1}
 B 0.25 ms^{-1}
 C 0.50 ms^{-1}
 D 1.0 ms^{-1}
 E $\frac{\pi}{2} \text{ ms}^{-1}$
 F 2.0 ms^{-1}

- 9 A uniform ladder rests in equilibrium with its upper end against a smooth vertical wall and its lower end on rough horizontal ground. The base of the ladder is at a horizontal distance of 3.0 m from the wall, and the top of the ladder is at a vertical height of 4.0 m. The ladder is on the point of slipping.



What is the coefficient of friction between the ladder and the ground?

- A $3/8$
 B $3/5$
 C $2/3$
 D $3/4$
 E $4/3$
- 10 A block of weight W rests on a rough plane inclined at 45° to the horizontal. The coefficient of static friction between the block and the plane is 0.5.

A horizontal force P is applied to the block.

Which inequality gives the range of values of P for which the block remains stationary?

- A $\frac{1}{2}W \leq P \leq \frac{3}{2}W$
 B $\frac{\sqrt{2}}{4}W \leq P \leq \frac{3\sqrt{2}}{4}W$
 C $\frac{1}{3}W \leq P \leq \frac{3}{2}W$
 D $\frac{1}{3}W \leq P \leq 3W$

- 11** A solid uniform cylinder of height 1.0 m and density 600 kg m^{-3} is held in a container of liquid of density 1000 kg m^{-3} by a light, vertical cable attached to the bottom of the container, as shown in the diagram.

In this position, the cylinder is upright with a depth of 0.90 m submerged below the surface of the liquid.

The cable is suddenly cut.

What is the initial acceleration of the cylinder immediately after the cable is cut?

(gravitational field strength = 10 N kg^{-1} ; assume resistive forces from the liquid are negligible at the instant of release)

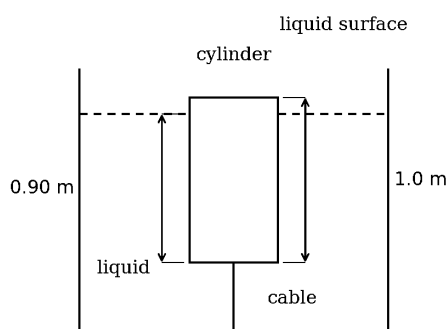


Diagram not to scale

- A** 1.0 m s^{-2}
B 3.0 m s^{-2}
C 4.0 m s^{-2}
D 5.0 m s^{-2}
E 6.7 m s^{-2}
F 9.0 m s^{-2}
- 12** An electric kettle has a power input of 2.5 kW. When used to boil water, it produces steam at a steady rate of 1.0 g s^{-1} .

What is the rate of energy loss from the kettle to the surroundings?

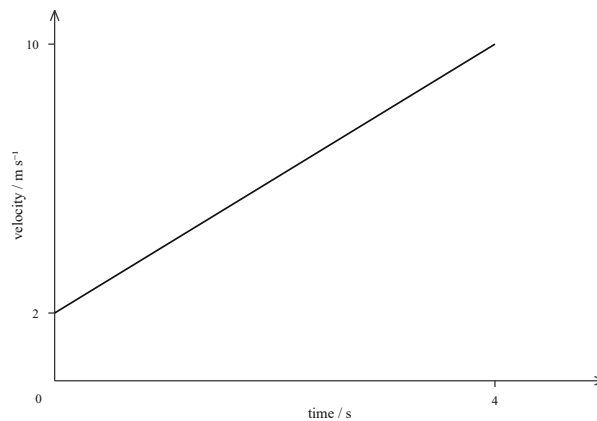
(Specific latent heat of vaporisation of water = $2.0 \times 10^6 \text{ J kg}^{-1}$; specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)

- A** 0 W
B 500 W
C 1500 W
D 2000 W
E 2500 W
F 4500 W

- 13** A nucleus of Thorium-232 (${}_{90}^{232}\text{Th}$) decays through a series of emissions to become a stable nucleus of Lead-208 (${}_{82}^{208}\text{Pb}$).
The decay series consists entirely of alpha (α) particles and beta-minus (β^-) particles.
What are the total numbers of α and β^- particles emitted during this process?
- A** 4 α , 6 β^-
B 6 α , 4 β^-
C 6 α , 8 β^-
D 6 α , 12 β^-
- 14** A small heating element at the bottom of an open tank of liquid heats a small parcel of the liquid immediately above it.
Just before this heated parcel begins to rise, how do its mass, weight, and the upthrust acting on it compare to their values before it was heated?
- A** Mass: Unchanged, Weight: Unchanged, Upthrust: Increases
B Mass: Unchanged, Weight: Unchanged, Upthrust: Decreases
C Mass: Unchanged, Weight: Decreases, Upthrust: Unchanged
D Mass: Decreases, Weight: Decreases, Upthrust: Unchanged
E Mass: Decreases, Weight: Decreases, Upthrust: Increases
F Mass: Unchanged, Weight: Decreases, Upthrust: Increases
- 15** Two identical copper spheres, **X** and **Y**, are heated to 500 K.
Sphere **X** has a matte black surface.
Sphere **Y** has a polished silver surface.
The initial rate of energy loss, R , is determined for each sphere in a **vacuum** and in an atmosphere of **argon** gas at 298 K.
In the vacuum, it is observed that $R_X > R_Y$.
Assume that the rate of heat transfer due to convection depends only on the dimensions of the sphere and the temperature difference.
Which statement correctly describes the rates in the argon atmosphere compared to the vacuum?
- A** The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ decreases.
B The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ stays constant.
C The rate R_X increases more than R_Y because black surfaces are better thermal conductors, and the ratio $\frac{R_X}{R_Y}$ increases.
D The rate R_Y increases more than R_X because the polished surface reflects gas molecules, and the ratio $\frac{R_X}{R_Y}$ decreases.

- 16** A continuous wave travels from medium X to medium Y. The wave speed in medium Y is $\frac{2}{3}$ of the wave speed in medium X.
What is the fractional change in the wavelength of the wave?
- A** $-\frac{1}{3}$
B 0
C $\frac{1}{3}$
D $\frac{1}{2}$
E $\frac{2}{3}$
- 17** The radioactive nuclide ${}_{92}^{235}\text{U}$ decays into the stable nuclide ${}_{82}^{207}\text{Pb}$ via a series of α (alpha) and β^- (beta-minus) emissions.
Which option gives the total number of α and β^- particles emitted during this decay chain?
- A** 5 α , 0 β^-
B 7 α , 4 β^-
C 7 α , 10 β^-
D 8 α , 6 β^-
- 18** An electric heater of constant power is used to heat a metal block. The block is initially at room temperature, 20 °C.
After 10 minutes, the temperature of the block is 50 °C.
The rate of thermal energy loss to the surroundings is proportional to the temperature difference between the block and the surroundings.
Which of the following describes the temperature θ of the block after a total of 20 minutes?
- A** $\theta = 50^\circ\text{C}$
B $50^\circ\text{C} < \theta < 80^\circ\text{C}$
C $\theta = 80^\circ\text{C}$
D $80^\circ\text{C} < \theta < 100^\circ\text{C}$
E $\theta = 100^\circ\text{C}$

- 19 An object of mass 2.0 kg moves in a straight line. The velocity-time graph for its motion is shown.



What is the magnitude of the constant net force acting on the object?

- A 1.0 N
 - B 2.0 N
 - C 4.0 N
 - D 5.0 N
 - E 12 N
 - F 20 N
- 20 A mechanical system with natural frequency f_0 is driven by a periodic force of constant amplitude and variable frequency f . The system is lightly damped, resulting in a sharp resonance peak in the amplitude A at $f \approx f_0$. Which of the following values of the driving frequency f corresponds to the condition where the rate of change of amplitude with respect to frequency, $\frac{dA}{df}$, is **positive** and has the **largest magnitude**?
- A $0.1f_0$
 - B $0.9f_0$
 - C $1.0f_0$
 - D $1.1f_0$
 - E $2.0f_0$

- 21** A cylindrical metal wire has an electrical resistance R . The wire is uniformly stretched until its length is 3 times its original length.

Assume that the volume of the wire remains constant and the resistivity of the metal does not change.

What is the resistance of the stretched wire?

- A** $\frac{R}{9}$
- B** $\frac{R}{3}$
- C** $3R$
- D** $9R$
- E** $27R$

- 22** Twelve identical resistors are connected to form the edges of a cube. A current enters the network at one vertex and leaves at the diagonally opposite vertex.

Let P_{in} be the power dissipated in a single resistor connected directly to the input vertex.

Let P_{mid} be the power dissipated in a single resistor that is connected to neither the input nor the output vertex.

What is the value of the ratio $\frac{P_{\text{in}}}{P_{\text{mid}}}$?

- A** 1
- B** 2
- C** 4
- D** 9
- E** 16

- 23** A solid composite object has total volume $4.0 \times 10^{-3} \text{ m}^3$ and is made from materials X and Y. Their densities are 2000 kg m^{-3} and 800 kg m^{-3} respectively.

It is fully submerged and neutrally buoyant in a fluid of density 1100 kg m^{-3} . For this question, neutral buoyancy means that the object's total mass equals the mass of the displaced fluid:

$$m_{\text{total}} = \rho_{\text{fluid}} V_{\text{total}}.$$

What is the mass of material X?

- A** 1.1 kg
- B** 2.0 kg
- C** 2.4 kg
- D** 4.0 kg
- E** 4.4 kg
- F** 6.0 kg

- 24** A radioactive source is monitored for a total of 60 s.

For an initial period of time, the measured count rate is $100 \text{ counts s}^{-1}$.

For the remainder of the time, the count rate is 40 counts s^{-1} .

The average count rate over the whole 60 s period is 60 counts s^{-1} .

What is the duration of the initial period of time?

- A** 12 s
- B** 20 s
- C** 24 s
- D** 30 s
- E** 36 s
- F** 40 s

- 25** A fixed mass of an ideal gas initially has pressure P and volume V . The gas is compressed at constant temperature to a final volume of $V/3$.

Which expression gives the final pressure of the gas?

- A** $\frac{P}{9}$
- B** $\frac{P}{3}$
- C** P
- D** $3P$
- E** $9P$

- 26** Two small spheres, X and Y, are initially held at the same height above the ground.
 At time $t = 0$, sphere X is released from rest and falls freely under gravity.
 At time $t = T$, sphere Y is projected vertically downwards from the same initial height with speed u .
- Both spheres fall freely under gravity of magnitude g , and air resistance is negligible. Assume that neither sphere has reached the ground.
- Which of the following statements about the motion of the two spheres for $t > T$ is/are correct?
- 1 The acceleration of sphere X relative to sphere Y is zero.
 - 2 If $u = gT$, the vertical separation between the two spheres remains constant.
 - 3 If $u = 2gT$, sphere Y catches up with sphere X at time $t = \frac{3}{2}T$.
- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3
- 27** A DC power supply of voltage V and negligible internal resistance is connected in series with two resistors of resistances R and $2R$.
 Which expression gives the current flowing through the resistor of resistance R ?
- A** $\frac{V}{3R}$
B $\frac{V}{2R}$
C $\frac{2V}{3R}$
D $\frac{V}{R}$
E $\frac{3V}{2R}$